

Estimating the Relationship Between Gasoline Prices and U.S. Gasoline Consumption: Evidence from 2010–2025

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BFIN 515 Economic Analysis

October 5, 2026

GitHub Repository URL: <https://github.com/tfrank-briggs/us-gasoline-demand-analysis>

1. Business Question

Gasoline is an important input for transportation in the United States, and changes in gasoline prices directly affect both consumers and businesses. Retail gasoline prices can fluctuate considerably because of changes in crude oil prices, refining conditions, seasonal demand, supply disruptions, taxes, and broader economic conditions (U.S. Energy Information Administration [EIA], n.d.-a). For businesses operating in the gasoline market, understanding how consumers respond to these price changes is important for evaluating pricing decisions, forecasting sales, and anticipating changes in market demand.

The central question of this study is: How is U.S. gasoline consumption associated with retail gasoline prices after accounting for real income and seasonality, and what do the results imply for businesses operating in the gasoline market?

Economic theory predicts an inverse relationship between the price of a good and the quantity demanded, holding other relevant factors constant. However, the size of this response can differ considerably across products. Gasoline provides an interesting case because many consumers rely on automobiles for commuting, shopping, and other transportation needs. Consumers may therefore have limited ability to immediately change their gasoline consumption when prices change. Over longer periods, however, consumers may have more opportunities to adjust their behavior by driving less, purchasing more fuel-efficient vehicles, using alternative transportation, or changing where they live and work.

This study uses historical U.S. gasoline price and consumption data to estimate their relationship. Price elasticity measures the percentage change in quantity demanded associated with a one-percent change in price, holding other demand determinants constant. The analysis

estimates a conditional percentage association and evaluates whether it can support a reliable elasticity interpretation, given the limitations of aggregate market data.

The results could be useful to fuel retailers, distributors, analysts, investors, and other decision makers interested in the gasoline market. Understanding how market prices correspond with sales volume can inform revenue and inventory planning. The full-sample price estimate is not statistically significant, however, so the strongest operational evidence from this model concerns recurring seasonal differences rather than a specific pricing rule.

2. Data

This study uses monthly U.S. data covering the period from January 2010 through December 2025. The dataset combines information on retail gasoline prices, gasoline consumption, and real disposable personal income. Using monthly observations provides sufficient variation over time to examine the relationship between gasoline prices and consumption while also allowing the analysis to account for seasonal changes in gasoline demand.

The primary measure of gasoline consumption is U.S. Product Supplied of Finished Motor Gasoline, obtained from the U.S. Energy Information Administration (EIA, n.d.-c). It is measured in thousand barrels per day as a monthly average daily rate, rather than the month's total volume. Product supplied measures petroleum leaving the primary supply chain and is commonly used as a proxy for domestic consumption; it does not directly measure final purchases by motorists (EIA, n.d.-b).

The main explanatory variable is the U.S. Regular All Formulations Retail Gasoline Price, also obtained from the EIA and measured in nominal dollars per gallon (EIA, n.d.-d). This national monthly series represents regular gasoline prices, while the quantity series covers

finished motor gasoline across grades. The coverage mismatch is a measurement limitation. Both EIA series are not seasonally adjusted.

The analysis also includes aggregate real disposable personal income as an economic control variable. The series is produced by the U.S. Bureau of Economic Analysis (BEA) and obtained through the Federal Reserve Bank of St. Louis FRED database as DSPIC96 (BEA, n.d.). It is reported monthly at a seasonally adjusted annual rate in billions of chained 2017 dollars. Income is included because changes in purchasing power may affect gasoline consumption independently of prices. The published series is retained without conversion to per-capita income or division by 12.

The final sample contains 192 unique, consecutive monthly observations, with complete and positive values for all three series. The workbook records an October 3, 2026 retrieval date. The series were joined by calendar month without interpolation or imputation. Eleven month-of-year indicators account for recurring seasonal patterns; January is the reference month.

Gasoline consumption, gasoline price, and real disposable personal income are transformed using natural logarithms. This transformation allows the price and income coefficients to describe percentage associations. Table 1 summarizes the original units. Figures 1 and 2 show the monthly price and consumption series, including the sharp consumption decline in April 2020 and the price peak in June 2022.

Table 1. Descriptive statistics for the monthly sample

Variable and units	Mean	Sample SD	Minimum	Maximum
Gasoline consumption (thousand barrels/day)	8,946.69	470.46	5,866.00	9,834.00
Gasoline price (nominal dollars/gallon)	3.010	0.592	1.764	4.929
Real disposable income (billions of chained 2017 dollars)	15,099.60	1,908.12	12,285.80	20,510.30

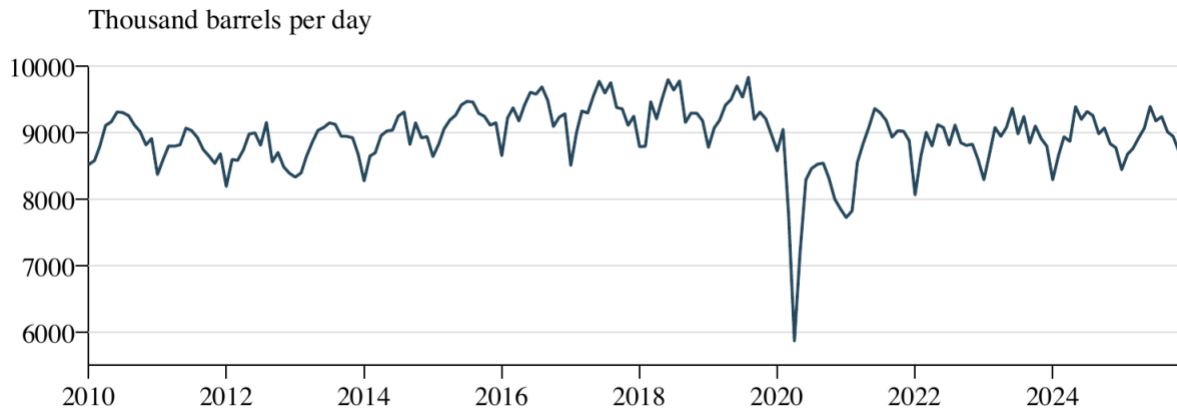
Source: US_Gasoline_Demand_2010_2025.xlsx, Monthly Data!B2:D193. All variables have $N = 192$. SD is the sample standard deviation ($N - 1$ denominator). Consumption is a monthly average daily rate; income is reported at a seasonally adjusted annual rate.

Figure 1. Monthly U.S. retail gasoline prices, 2010–2025



Source: supplied workbook, Monthly Data!C2:C193. Prices are nominal and not seasonally adjusted.

Figure 2. Monthly U.S. gasoline consumption proxy, 2010–2025



Source: supplied workbook, Monthly Data!B2:B193. Product supplied is a consumption proxy and is not seasonally adjusted.

3. Method

This study uses multiple regression analysis to estimate the relationship between retail gasoline prices and U.S. gasoline consumption while controlling for changes in real income and recurring seasonal patterns. The model estimates a conditional association; interpreting its price coefficient as a causal demand elasticity would require additional identification assumptions.

The baseline model is specified as:

$$\ln(Q_t) = \beta_0 + \beta_1 \ln(P_t) + \beta_2 \ln(I_t) + \sum_{m=2}^{12} \delta_m D_{m,t} + \varepsilon_t$$

Here, Q_t is the gasoline consumption proxy, P_t is nominal retail gasoline price, and I_t is aggregate real disposable personal income in month t . $D_{m,t}$ equals 1 when observation t occurs in calendar month m and 0 otherwise, with $m = 2$ for February through $m = 12$ for December. January is omitted, so every month coefficient δ_m compares that month with January.

β_0 is the intercept, β_1 is the conditional price coefficient, β_2 is the conditional income coefficient, and ε_t is the error term. Including all 12 month indicators alongside an intercept would create perfect multicollinearity; January remains in the sample with all 11 indicators equal to zero.

The natural-log specification makes the coefficients easier to interpret in percentage terms. The price coefficient, β_1 , represents the approximate percentage difference in gasoline consumption associated with a 1% difference in retail gasoline price, holding income and calendar month constant.

For example, if β_1 were estimated to be -0.25 , a 1% higher gasoline price would be associated with approximately 0.25% lower consumption, holding the other variables constant. This is the usual log-log interpretation, but the observed association alone does not identify a movement along a demand curve.

Economic theory predicts a negative price effect on quantity demanded when other demand determinants are held constant. A causal elasticity with an absolute value below one would indicate price-inelastic demand, while an absolute value above one would indicate price-elastic demand. The present model does not by itself establish such a causal elasticity or distinguish a short-run response from a long-run response.

Real income controls for purchasing power, which may influence travel and gasoline consumption independently of prices. The month indicators account for recurring seasonal differences, including summer travel patterns. These controls improve comparability across observations but cannot capture every factor affecting consumption.

The model was estimated using ordinary least squares (OLS), which minimizes the sum of squared residuals. The specification contains an intercept and 13 explanatory variables: log price, log income, and 11 month indicators. With 192 observations, the baseline regression has 178 residual degrees of freedom. Descriptive statistics and trends were examined before interpreting the direction, magnitude, and significance of the estimated price coefficient.

The estimated relationship should be interpreted with caution. Gasoline prices and consumption both reflect market conditions, including simultaneous supply and demand changes. Dong (2026) explains that aggregate prices and quantities require instruments or other identifying restrictions to separate supply and demand functions. This regression includes neither instruments nor a structural identification strategy. It also uses nominal gasoline prices with real income and omits a time trend, dynamic adjustments, and other possible demand determinants. Monthly indicators address recurring seasonality, while the validation checks address uncertainty and sample sensitivity; neither resolves endogeneity.

4. Results

The analysis uses 192 monthly observations covering January 2010 through December 2025. During this period, average U.S. gasoline consumption was approximately 8,946.69 thousand barrels per day, with a minimum of 5,866 thousand barrels per day and a maximum of 9,834 thousand barrels per day. The average retail gasoline price was approximately \$3.01 per gallon, ranging from \$1.764 to \$4.929 per gallon. These variations provide an opportunity to examine how gasoline consumption changed alongside gasoline prices over the sample period.

The regression produced an R-squared of 0.2589 and an adjusted R-squared of 0.2047. Approximately 25.9% of the sample variation in log consumption is explained by gasoline prices, real income, and seasonal controls. Under conventional OLS assumptions, the regressors are jointly significant: $F(13, 178) = 4.7822$, with $p = 4.18 \times 10^{-7}$. This test does not establish that price alone is significant or that the model predicts accurately outside the sample. Because residual autocorrelation is substantial, the conventional F test should be interpreted alongside the validation discussion; the workbook does not report a HAC-adjusted joint F test.

The main coefficient of interest is the coefficient on log gasoline price. The estimate is -0.0094 , with a conventional standard error of 0.0184, $t = -0.511$, and $p = 0.6103$. It suggests that a 1% higher gasoline price is associated with approximately 0.0094% lower consumption, holding real income and calendar month constant. Table 2 reports all baseline coefficients and model statistics.

Table 2. Baseline OLS regression for log gasoline consumption

Term	Coefficient	Standard error	t statistic	p value
Intercept	9.5650	0.2795	34.218	3.36e-80
ln(Price)	-0.0094	0.0184	-0.511	0.6103
ln(Income)	-0.0540	0.0289	-1.866	0.0638
February	0.0344	0.0177	1.938	0.0542
March	0.0546	0.0178	3.071	0.0025
April	0.0434	0.0178	2.435	0.0159
May	0.0760	0.0179	4.256	3.36e-05
June	0.0995	0.0179	5.573	9.12e-08
July	0.0925	0.0178	5.183	5.88e-07
August	0.1021	0.0178	5.728	4.25e-08
September	0.0682	0.0178	3.827	0.0002
October	0.0692	0.0178	3.894	0.0001
November	0.0543	0.0177	3.059	0.0026
December	0.0487	0.0177	2.746	0.0067
N = 192; residual degrees of freedom = 178				
R ² = 0.2589; adjusted R ² = 0.2047				
F(13, 178) = 4.7822; F-test p value = 4.18e-07				

Source: supplied workbook, OLS Results!A6:E20 and A23:B30. January is the omitted reference month. Coefficients use natural logarithms of consumption, price and income. Conventional OLS standard errors and two-sided t tests assume homoskedastic, uncorrelated errors. Robust inference appears in Table 3. Scientific notation preserves very small p values.

Although the negative sign is consistent with the expected demand response, the price coefficient is not statistically significant at conventional levels. The full-sample model therefore does not provide sufficient evidence of a systematic conditional price association. A large p value does not establish that price has no effect. Although the point estimate has a magnitude below one, it would be inappropriate to conclude that gasoline demand is definitively price

inelastic because the coefficient is imprecise and the model does not identify a causal demand response.

The real income coefficient is also insignificant at the 5% level: -0.0540 , with a conventional p value of 0.0638 . Its negative sign differs from the usual expectation for a normal good. Given the uncertainty and the model's limitations, this should not be interpreted as evidence that higher income reduces gasoline demand.

Seasonal patterns were clearer. Relative to January, 10 of the 11 month coefficients are positive and significant at the 5% level under conventional OLS inference; February is the exception ($p = 0.0542$). The largest estimated differences occur in August (0.1021), June (0.0995), and July (0.0925). These are differences in log consumption, rather than elasticities. The pattern is consistent with higher gasoline consumption during summer travel months, although the regression does not directly measure travel behavior.

The baseline model provides clearer evidence of recurring seasonality than of a stable price relationship. Its limited explanatory power and imprecise price estimate also leave room for additional factors and changes in market conditions to explain consumption over time.

5. Validation

To evaluate the reliability of the baseline results, additional checks were performed to determine whether the estimated relationship between gasoline prices and consumption was sensitive to the statistical assumptions and sample period used in the original model.

The first validation check addresses residual autocorrelation. Monthly observations close together in time may be related, violating the conventional OLS assumption of uncorrelated errors. The workbook reports a Durbin–Watson statistic of 0.487 , suggesting substantial positive

residual correlation. This can make conventional standard errors and significance tests unreliable even when the coefficient estimates remain unchanged.

Heteroskedasticity and autocorrelation consistent (HAC) standard errors with 12 monthly lags were therefore examined. The workbook uses a Newey–West covariance calculation with Bartlett weights and an $N/(N - K)$ finite-sample correction, where K counts estimated coefficients including the intercept. HAC adjusts uncertainty while retaining the OLS coefficients; the official statsmodels documentation describes this class of estimator (statsmodels developers, n.d.). Robust p values use two-sided asymptotic normal tests, while conventional OLS uses t tests. The price standard error rises from 0.0184 to 0.0730 and its p value rises to 0.8976. Its 95% HAC interval is $[-0.1525, 0.1337]$. Income also remains insignificant (HAC standard error = 0.0633; $p = 0.3936$). Table 3 summarizes the checks.

Table 3. Price coefficient sensitivity and robust inference

Specification	N	Price coefficient	Price SE	Price p value	R ²
Full sample, OLS	192	-0.0094	0.0184	0.6103	0.2589
Full sample, HAC(12)	192	-0.0094	0.0730	0.8976	0.2589
Full sample, HAC(6)	192	-0.0094	0.0637	0.8827	0.2589
Full sample, HAC(3)	192	-0.0094	0.0554	0.8652	0.2589
Full sample, HC1	192	-0.0094	0.0351	0.7888	0.2589
Pandemic indicator, HAC(12)	192	-0.0510	0.0428	0.2337	0.4893
Exclude Mar 2020–Dec 2021, HAC(12)	170	-0.1187	0.0114	1.59e-25	0.8517
Exclude Mar–Dec 2020, HAC(12)	182	-0.1088	0.0148	2.16e-13	0.7549
Exclude Mar 2020–Jun 2021, HAC(12)	176	-0.1178	0.0115	1.01e-24	0.8471

Source: supplied workbook, Robustness!A5:L14. All models include log income and February–December controls. The pandemic indicator equals 1 for March 2020–December 2021. HAC and HC1 p values use two-sided asymptotic normal tests; OLS uses a t distribution. R² comparisons across different samples are descriptive.

A second validation check examines sensitivity to the COVID-19 period. EIA (2020) documents reduced transportation fuel consumption during stay-at-home orders and travel limitations. In the workbook, consumption reaches its minimum of 5,866 thousand barrels per day in April 2020, compared with the sample average of 8,946.69. The model does not separately identify the effects of travel restrictions, remote work, or other pandemic-related changes.

Excluding March 2020–December 2021 removes 22 months and leaves $N = 170$. The price coefficient becomes -0.1187 (HAC standard error = 0.0114; $p = 1.59 \times 10^{-25}$), with a 95% interval of $[-0.1410, -0.0964]$. Excluding March–December 2020 yields -0.1088 ($N = 182$), while excluding March 2020–June 2021 yields -0.1178 ($N = 176$); both are significant with HAC(12) inference. Adding a March 2020–December 2021 indicator instead retains all 192 observations and produces -0.0510 ($p = 0.2337$). These comparisons show substantial sample sensitivity; they do not establish a single causal elasticity applicable throughout 2010–2025.

The full sample remains the primary analysis. HAC corrections leave its price coefficient insignificant, while pandemic exclusions reveal a stronger negative association. These exploratory comparisons are not adjusted for multiple testing, and their higher R-squared values are calculated on different samples. For excluded observations, the workbook preserves original calendar-month lag distances rather than treating retained months on either side of a gap as adjacent. HAC(3), HAC(6), and heteroskedasticity-only HC1 checks also leave the full-sample price coefficient insignificant. Seasonal evidence persists under HAC(12), although April loses significance and February becomes significant; 10 of the 11 month contrasts remain significant at 5%. Robust standard errors do not address endogeneity, nonstationarity, or omitted-variable bias, and the workbook reports no stationarity or cointegration tests.

6. Recommendation

The results of this study suggest that businesses operating in the U.S. gasoline market should be cautious about relying on a single historical price-elasticity estimate when making pricing, sales, or inventory decisions. Although the estimated gasoline price coefficient was negative, as predicted by economic theory, the relationship was not statistically significant in the baseline model and was sensitive to changes in the sample period. Therefore, the analysis does not provide sufficient evidence to recommend a specific price increase or decrease based solely on the estimated elasticity.

For fuel retailers and distributors, one of the clearest findings is the importance of seasonality. Gasoline consumption was significantly higher during most months relative to January, with particularly strong effects during the summer months. This suggests that businesses should incorporate seasonal demand patterns into inventory planning, staffing, purchasing, and sales forecasts. Preparing for higher gasoline consumption during periods such as June, July, and August may be more reliable than assuming that a particular percentage change in price will produce a predictable percentage change in sales volume.

Businesses should also consider broader market conditions when forecasting gasoline demand. The COVID-19 period illustrates how changes in travel and economic activity may affect consumption alongside price changes. The sensitivity of the price coefficient to pandemic exclusions shows that estimates depend on the sample period, but it does not separately identify changes in consumer behavior.

For pricing decisions, the results should therefore be treated as one source of information rather than a complete pricing rule. Individual gasoline retailers may benefit from combining national market data with more detailed information on local competitors, wholesale fuel costs,

traffic patterns, customer characteristics, location, and historical station-level sales. Such data could provide a more accurate estimate of the demand conditions facing a particular retailer than national aggregate data alone.

The main recommendation is to combine price information, seasonal patterns, and current market conditions when forecasting sales and making operational decisions. Historical national prices alone do not provide a reliable pricing rule in this analysis. Any forecast based on these results should be validated against subsequent sales data.

7. Conclusion

This study examined the relationship between retail gasoline prices and gasoline consumption in the United States using 192 monthly observations from January 2010 through December 2025. The objective was to determine how responsive gasoline consumption is to changes in retail prices and to consider how this information could assist businesses operating in the gasoline market.

The baseline gasoline price coefficient is -0.0094 . Although its sign is consistent with the expected demand response, it is not statistically significant under conventional OLS or HAC(12) inference. The full-sample analysis therefore does not establish a statistically distinguishable conditional price association after controlling for real disposable personal income and seasonal effects.

Seasonality produces a clearer pattern, with higher consumption during most months relative to January and the largest differences in summer. These findings support considering recurring seasonal patterns in operational planning, subject to validation with local sales data.

The validation analysis demonstrates the limitations of interpreting the baseline coefficient as a stable demand elasticity. HAC(12) raises the price p value to 0.8976, while

excluding March 2020–December 2021 produces a significant negative association of -0.1187 . The difference shows that the estimated association is sensitive to the period analyzed. Neither finding, on its own, identifies a causal price response.

These findings reflect the broader challenge of estimating supply and demand relationships from aggregate market data. Observed prices and quantities arise from both supply and demand, making a causal demand response difficult to isolate without additional identifying information or assumptions (Dong, 2026). The results should therefore be interpreted as evidence about conditional associations between gasoline prices and consumption.

For businesses in the gasoline market, pricing decisions and sales forecasts should not rely on a single national price-elasticity estimate. Seasonal patterns, local market conditions, wholesale costs, and unusual economic disruptions should also be considered. Future analysis using more detailed geographic or retailer-level data, together with a credible identification strategy, could better assess how consumers respond to gasoline price changes.

Data and Code Availability

The analysis uses public EIA and BEA/FRED monthly series for January 2010–December 2025. The accompanying workbook, `US_Gasoline_Demand_2010_2025.xlsx`, contains the 192 observations, source links and retrieval notes, log and month-indicator formulas, baseline OLS estimates, and robust sensitivity results. Its numerical results are the authoritative source for the tables and figures in this paper. Regression results are saved estimates and must be recomputed after input changes. Data and analysis code are intended to be deposited in a GitHub repository; no public repository URL is available at submission.

GitHub Repository URL: [INSERT REPOSITORY URL]

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